Evaluating the Effectiveness of Self-Correction Prompting Strategies in Mitigating LLM Hallucinations

1. Project Title

Evaluating the Effectiveness of Self-Correction Prompting Strategies in Mitigating LLM Hallucinations

2. Team Members & Responsibilities

Wang Shaoyu (, 2311018) - Project Lead: Overall coordination, literature review, experiment design, implementation, report writing

3. Background & Motivation

Large Language Models (LLMs) have demonstrated remarkable capabilities across various natural language processing tasks. However, these models are prone to "hallucination" - generating content that is fluent, coherent, but factually incorrect, unverifiable, or contradictory to source information. This phenomenon poses significant challenges for deploying LLMs in knowledge-intensive and high-stakes applications such as medical diagnosis, legal consultation, and news generation.

Various methods have been proposed to mitigate hallucination, including retrieval-augmented generation, model fine-tuning, and prompting strategies. Among these, prompting-based approaches are particularly attractive due to their accessibility and low computational cost. Chain-of-Thought (CoT) prompting and self-reflection prompting are two promising directions that encourage models to reason step by step or explicitly verify their outputs.

Despite the growing interest, there lacks a systematic evaluation of whether these prompting strategies genuinely reduce hallucination across different tasks and models, and under what conditions they are effective.

4. Research Questions

RQ1: To what extent can Chain-of-Thought prompting reduce hallucination in LLM-generated text?

RQ2: How effective are self-reflection and self-correction prompting strategies in detecting and mitigating hallucination?

RQ3: What are the limitations of prompting-based approaches for hallucination mitigation? When do they fail?

5. Methodology

5.1 Tasks and Datasets

- Factuality evaluation using TruthfulQA
- Summarization faithfulness evaluation using CNN/DailyMail
- Open-domain question answering (e.g., Natural Questions)

5.2 Models

- **GPT-2** (124M parameters) - primary baseline model
- **Qwen2-1.5B** - enhanced comparison model
- (Optional) Claude/GPT-3.5 via API for broader evaluation

Note: Experiments will be conducted on a laptop with RTX 4050 (6GB VRAM), enabling efficient local inference.

5.3 Prompting Strategies (Comparison)

1. Standard prompting (baseline)
2. Chain-of-Thought (CoT) prompting
3. Self-reflection prompting (with explicit verification step)
4. Self-consistency with multiple samples

5.4 Evaluation Metrics

- Accuracy / F1 on factuality benchmarks
- FActScore-style fine-grained factual evaluation
- Human evaluation (if time permits)

5.5 Experimental Plan

- Design 4-6 carefully crafted prompts per strategy
- Generate responses for each dataset
- Evaluate using automated metrics
- Analyze failure cases and error patterns

6. Expected Outcomes

- A comparative analysis of prompting strategies for hallucination mitigation
- Experimental results showing effectiveness of CoT and self-reflection methods
- Identification of conditions under which prompting strategies succeed or fail
- An 8-page ACL-format research report in English

7. Project Timeline

Phase	Period	Tasks
Phase 1	Oct - Nov 2026	Complete literature review on hallucination and prompting; finalize experimental design and prompts; set up evaluation pipeline
Phase 2	Nov - Mid Dec 2026	Run experiments on all models and datasets; collect and analyze results; submit midterm milestone
Phase 3	Mid Dec - Late Dec 2026	Complete data analysis; write and polish final report; prepare presentation slides

8. References

[1] Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring how models mimic human falsehoods. *ACL 2022*.

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[3] Wei, J., Wang, X., Schuurmans, D., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS 2022*.

[4] Manakul, P., Liusie, A., & Gales, M. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. *EMNLP 2023*.

[5] Ji, Z., Lee, N., Frieske, R., et al. (2023). Survey on hallucination in LLMs. *Findings of ACL 2023*.

[6] Madaan, A., Tandon, N., Gupta, P., et al. (2023). Self-refine: Iterative refinement with self-feedback. *NeurIPS 2023*.